

COURSE CODE:CSA16

COURSE NAME: Data Warehousing and Data Mining

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SCENARIO BASED QUESTIONS

Q1. Data Preprocessing in Retail Analytics (*L3, L4*)

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

- Identify missing values in **Age**, **Monthly_Spend**, and **Purchase_Frequency**
- Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
- Compute the imputed values and justify the selected method

B. Outlier Detection and Treatment

- Detect outliers in **Age** and **Monthly_Spend** using the **Z-score method**
- Identify records containing extreme values (e.g., Age = 105, Spend = 500)
- Apply suitable treatment methods (**capping / transformation / removal**) and justify

C. Categorical Data Standardization

- Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)

ii. Convert all entries into a standardized format (**Male / Female**)

SOLUTION:

A. Handling Missing Values

i. Missing values identified

Field	Missing in
Age	C2
Monthly_Spend	C3
Purchase_Frequency	C5

ii & iii. Imputation — Method 1: Median Imputation

Median is preferred over mean here because the dataset contains extreme outliers (C4: Age=105, Spend=500), and mean is heavily pulled by outliers while median is robust to them.

- **Age** (excluding C2): 23, 27, 31, 36, 40, 42, 105 → sorted → median = **36** → impute C2 Age = 36
- **Monthly_Spend** (excluding C3): 20, 25, 35, 38, 45, 300, 500 → sorted → median = **38** → impute C3 Spend = 38
 - (Note: mean here = 137.6 — clearly distorted by 300 and 500, confirming median is the right choice)
- **Purchase_Frequency** (excluding C5): 5, 8, 9, 10, 12, 45, 50 → sorted → median = **10** → impute C5 Frequency = 10

Imputation — Method 2: k-NN Imputation (k=3)

Using Euclidean distance on the other numeric attributes to find the 3 most similar customers:

Missing value	Nearest 3 neighbours (by profile similarity)	k-NN imputed value
C2 Age	C7(36), C6(40), C1(23)	$(36+40+23)/3 = \mathbf{33}$
C3 Spend	C7(38), C6(45), C1(20)	$(38+45+20)/3 = \mathbf{34.3}$
C5 Frequency	C1(5), C7(9), C6(12)	$(5+9+12)/3 = \mathbf{8.7}$

Justification: Median imputation is simple and outlier-resistant — good as a fast baseline. k-NN imputation is more contextually accurate because it borrows values from customers with similar spend/frequency/loyalty profiles rather than blending the whole population — e.g., C5 (Low loyalty) gets a value pulled toward other low-spend customers (8.7) rather than the population median (10), which better reflects its actual segment.

B. Outlier Detection and Treatment (Z-score method)

$$Z = (X - \mu) / \sigma$$

Age (using median-imputed dataset: 23,36,31,105,27,40,36,42):

- Mean = 42.5, Std $\approx$ 24.4 (population)

- $Z(C4, \text{Age}=105) = (105 - 42.5)/24.4 \approx 2.56 \rightarrow$ flagged outlier

Monthly_Spend (23,35,38,500,25,45,38,300 $\rightarrow$ using imputed 38 for C3):

- Mean = 125.1, Std $\approx$ 166.6
- $Z(C4, \text{Spend}=500) = (500 - 125.1)/166.6 \approx 2.25 \rightarrow$ flagged outlier
- $Z(C8, \text{Spend}=300) \approx 1.05 \rightarrow$ within normal range, not extreme

Extreme record identified: C4 (Age=105, Spend=500) is the clear outlier in both dimensions — likely a manual data-entry error (age 105 is implausible for an active shopper, and the spend is disproportionate to the rest of the sample).

Treatment recommendation:

- **Capping/Winsorization:** cap Age at a realistic business maximum (e.g., 70) and cap Spend at the 95th/99th percentile or a domain-defined ceiling — preserves the record instead of discarding it, useful for small datasets.
- **Alternative:** flag C4 for manual verification (105 may be a typo for 45 or 25) before deciding between correction, capping, or removal.
- **Log transformation** of Monthly_Spend is a good general option if outliers are frequent/systemic rather than a single erroneous point (they aren't here — only C4 is extreme).

C. Categorical Standardization

Original	Standardized
M	Male
Male	Male
F	Female
Female	Female
male	Male
FEMALE	Female
M	Male
Female	Female

Standardization rule: convert to lowercase $\rightarrow$ capitalize first letter $\rightarrow$ map single-letter codes (M $\rightarrow$ Male, F $\rightarrow$ Female).

Q3. Data Transformation & Discretization in Education Analytics

A. Equal-Width Binning (3 bins)

Study_Hours (min=2, max=11, range=9, **bin width** = $9/3 = 3$):

Bin	Interval	Students
Low	[2, 5)	S1(2), S2(3), S3(4)
Medium	[5, 8)	S4(5), S5(6), S6(7)
High	[8, 11]	S7(8), S8(9), S9(10), S10(11)

Test_Score (min=40, max=100, range=60, **bin width** = $60/3 = 20$):

Bin Interval Students

Poor [40, 60) S1(40), S2(50), S3(55)

Average [60, 80) S4(65), S5(75)

Excellent [80, 100] S6(85), S7(90), S8(95), S9(98), S10(100)

B. Equal-Frequency Binning (3 bins, ~equal record counts: 4-3-3)

Bin Study_Hours Test_Score

Bin 1 (S1–S4) 2, 3, 4, 5 40, 50, 55, 65

Bin 2 (S5–S7) 6, 7, 8 75, 85, 90

Bin 3 (S8–S10) 9, 10, 11 95, 98, 100

Labels assigned the same way: Bin1 = Low/Poor, Bin2 = Medium/Average, Bin3 = High/Excellent.

C. Concept Hierarchy

- **Study_Hours:** Low (2–5 hrs) → Medium (6–8 hrs) → High (9–11 hrs)
- **Test_Score:** Poor (40–59) → Average (60–79) → Excellent (80–100)

This hierarchy converts continuous values into interpretable categories aligned with the "Performance" labels already in the dataset, improving decision-making transparency for non-technical stakeholders.

Q4. Normalize: 200, 300, 400, 600, 1000

(a) Min-Max Normalization [0, 1]

$X' = (X - X_{\min}) / (X_{\max} - X_{\min})$, $X_{\min}=200$, $X_{\max}=1000$, range=800

X X'

200 $(200-200)/800 = 0$

300 $(300-200)/800 = 0.125$

400 $(400-200)/800 = 0.25$

600 $(600-200)/800 = 0.5$

1000 $(1000-200)/800 = 1.0$

(b) Z-score Normalization

Mean = $(200+300+400+600+1000)/5 = 500$

Variance (population) = $[(300^2+200^2+100^2+100^2+500^2)] / 5 = 400000/5 = 80000$

Std = $\sqrt{80000} \approx 282.84$

X Z = (X-500)/282.84

200 **-1.061**

300 **-0.707**

400 **-0.354**

600 +0.354

1000 +1.768

Q5. Data Transformation in Online Shopping Behavior Analysis

Data: 15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600

A. Equal-Frequency Binning (3 bins, sizes 4-3-3)

Bin Values

Bin 1 15, 30, 45, 60

Bin 2 120, 300, 600

Bin 3 900, 1800, 3600

Analysis: This groups quick page-bounces (Bin 1) separately from moderately engaged visits (Bin 2) and highly engaged sessions (Bin 3). It handles the skew reasonably well for classification purposes, but Bin 3 still spans a 4x range (900–3600s), so granularity at the high end is lost — equal-frequency binning prioritizes equal *sample counts* over equal *value ranges*.

B. Z-Score Normalization

Mean = $7470/10 = 747$

Sum of squared deviations = 11,901,060 → Variance = 1,190,106 → Std ≈ 1090.92

X Z = (X-747)/1090.92

15 -0.671

30 -0.657

45 -0.644

60 -0.630

120 -0.575

300 -0.410

600 -0.135

900 +0.140

1800 +0.965

3600 +2.616

Handling extreme values: The 3600s visit produces the largest Z-score (+2.62), correctly flagging it as an outlier relative to the distribution, but z-score does **not bound or cap** it — extreme values remain visibly extreme, which can still skew distance-based clustering algorithms even after normalization.

C. Min-Max Normalization [0, 1]

Xmin=15, Xmax=3600, range=3585. $X' = (X-15)/3585$

X X'

15 0.000

30 0.004

45 0.008

60 0.013

120 0.029

300	0.080
600	0.163
900	0.247
1800	0.498
3600	1.000

Comparison: Min-Max normalization is *more sensitive to the outlier* than Z-score here — because the single extreme value (3600) anchors the entire scale, most other points (even a substantial 600-second visit) get compressed close to 0 (0.163). Z-score instead expresses each value relative to the mean and spread of the *whole distribution*, giving a more balanced relative picture, though it too is influenced by the outlier's effect on mean/std. For this heavily right-skewed data, a **log-transform prior to normalization** would produce a more balanced result than either method alone.

Q1. Data Preprocessing in Retail Analytics

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i. Missing values identified

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Purchase_Frequency	C5

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 - (Note: mean here = 137.6 — clearly distorted by 300 and 500, confirming median is the right choice)
- **Purchase_Frequency** (excluding C5): 5, 8, 9, 10, 12, 45, 50 → sorted → median = **10** → impute C5 Frequency = 10

Imputation — Method 2: k-NN Imputation (k=3)

Using Euclidean distance on the other numeric attributes to find the 3 most similar customers:

Missing value Nearest 3 neighbors (by profile similarity) k-NN imputed value

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customers (8.7) rather than the population median (10), which better reflects its actual segment.

B. Outlier Detection and Treatment (Z-score method)

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- **Z(C4, Age=105) = (105 – 42.5)/24.4 $\approx$ 2.56** → flagged outlier

Monthly_Spend (23,35,38,500,25,45,38,300 → using imputed 38 for C3):

- Mean = 125.1, Std $\approx$ 166.6
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- Z(C8, Spend=300) $\approx$ 1.05 → within normal range, not extreme

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Treatment recommendation:

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Standardization rule: convert to lowercase → capitalize first letter → map single-letter codes (M→Male, F→Female).

Q2. Covariance & Data Reduction in E-commerce Analytics

A. Covariance Analysis (Income vs Purchase_Value)

Income: 40,60,85,120,45,75,150,70 → **Mean = 80.625**

Purchase_Value: 100,150,210,320,110,180,360,160 → **Mean = 198.75**

ID (Income – 80.625) (PV – 198.75) Product

U1 –40.625	–98.75	4011.7
U2 –20.625	–48.75	1005.5
U3 4.375	11.25	49.2
U4 39.375	121.25	4774.2
U5 –35.625	–88.75	3161.7
U6 –5.625	–18.75	105.5
U7 69.375	161.25	11185.5
U8 –10.625	–38.75	411.7

Sum of products ≈ **24705.1**

Cov(Income, Purchase_Value) = 24705.1 / 8 ≈ 3088.1 (population) — or 24705.1/7 ≈ 3529.3 (sample)

ii. Interpretation: Strongly **positive covariance** — Income and Purchase_Value increase together (higher-income customers spend more).

iii. Role of covariance: Covariance quantifies whether two variables move together (positive), oppositely (negative), or independently (near zero). It's a first step in detecting **redundant features** — highly correlated attributes carry overlapping information, so identifying them (via covariance/correlation) allows the removal of features that would otherwise inflate model complexity and cause multicollinearity.

B. Feature Selection

Across the dataset, Age, Income, Purchase_Value, Credit_Score, Orders, Avg_Balance, and EMI all rise together monotonically — indicating **strong mutual correlation and redundancy**.

Redundant attributes identified:

- **EMI** and **Avg_Balance** are largely proxies for **Income** (higher income → higher EMI eligibility → higher balance)
- **Orders** closely tracks **Purchase_Value** (more orders drives higher total spend)

Optimal subset selected: Income, Purchase_Value, Credit_Score, Orders

Justification: This subset retains features that capture *distinct dimensions* of customer behavior — financial capacity (Income), spending behavior (Purchase_Value), creditworthiness/risk (Credit_Score), and engagement frequency (Orders) — while dropping Age (weaker independent signal here), EMI, and Avg_Balance, which are largely linear proxies of Income. This reduces multicollinearity, lowers computational cost, and reduces overfitting risk without losing predictive information.

C. Data Transformation (Standardization)

Z = (X – μ) / σ, applied per column (excluding Cust_ID, Customer_Class)

Worked example — **Income** column:

- Mean = 80.625, Std ≈ 35.13
- U1: (40 – 80.625)/35.13 = **–1.156**
- U7: (150 – 80.625)/35.13 = **+1.974**

The same formula (compute column mean and std, then transform each value) is applied identically to Age, Purchase_Value, Credit_Score, Orders, Avg_Balance, and EMI.

Why standardize: The features are on very different scales — Age (tens), Credit_Score (hundreds), Income/Purchase_Value (tens to hundreds of thousands), EMI (thousands). Without standardization, large-magnitude features (EMI, Purchase_Value) would dominate distance-based models (k-NN, clustering) or slow gradient-based convergence, purely due to scale — not actual importance. Standardizing to mean 0, std 1 ensures every feature contributes fairly.

Q2. Covariance & Data Reduction in E-commerce Analytics (L4, L5)

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase Value (₹K)	Credit Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

- i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
- ii. Interpret whether the relationship is **positive or negative**

iii. Explain how covariance helps in identifying feature relationships

B. Feature Selection

- i. Identify redundant attributes using **correlation or domain knowledge**
 - ii. Select an optimal subset of features for modeling
 - iii. Justify your selection
-

C. Data Transformation

- i. Standardize the dataset (excluding **Cust_ID** and **Customer_Class**)
 - ii. Explain why normalization/standardization is required
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SOLUTION:

Q2. Covariance & Data Reduction in E-commerce Analytics

A. Covariance Analysis (Income vs Purchase_Value)

Income: 40,60,85,120,45,75,150,70 → **Mean = 80.625**

Purchase_Value: 100,150,210,320,110,180,360,160 → **Mean = 198.75**

ID	(Income – 80.625)	(PV – 198.75)	Product
U1	–40.625	–98.75	4011.7
U2	–20.625	–48.75	1005.5
U3	4.375	11.25	49.2
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Sum of products $\approx$ **24705.1**

Cov(Income, Purchase_Value) = $24705.1 / 8 \approx 3088.1$ (population) — or $24705.1/7 \approx 3529.3$ (sample)

ii. Interpretation: Strongly **positive covariance** — Income and Purchase_Value increase together (higher-income customers spend more).

iii. Role of covariance: Covariance quantifies whether two variables move together (positive), oppositely (negative), or independently (near zero). It's a first step in detecting **redundant features** — highly correlated attributes carry overlapping information, so identifying them (via covariance/correlation) allows the removal of features that would otherwise inflate model complexity and cause multicollinearity.

B. Feature Selection

Across the dataset, Age, Income, Purchase_Value, Credit_Score, Orders, Avg_Balance, and EMI all rise together monotonically — indicating **strong mutual correlation and redundancy**.

Redundant attributes identified:

- **EMI** and **Avg_Balance** are largely proxies for **Income** (higher income → higher EMI eligibility → higher balance)
- **Orders** closely tracks **Purchase_Value** (more orders drives higher total spend)

Optimal subset selected: Income, Purchase_Value, Credit_Score, Orders

Justification: This subset retains features that capture *distinct dimensions* of customer behavior — financial capacity (Income), spending behavior (Purchase_Value), creditworthiness/risk (Credit_Score), and engagement frequency (Orders) — while dropping Age (weaker independent signal here), EMI, and Avg_Balance, which are largely linear proxies of Income. This reduces multicollinearity, lowers computational cost, and reduces overfitting risk without losing predictive information.

C. Data Transformation (Standardization)

$Z = (X - \mu) / \sigma$, applied per column (excluding Cust_ID, Customer_Class)

Worked example — **Income** column:

- Mean = 80.625, Std $\approx$ 35.13
- U1: $(40 - 80.625)/35.13 = -1.156$
- U7: $(150 - 80.625)/35.13 = +1.974$

The same formula (compute column mean and std, then transform each value) is applied identically to Age, Purchase_Value, Credit_Score, Orders, Avg_Balance, and EMI.

Why standardize: The features are on very different scales — Age (tens), Credit_Score (hundreds), Income/Purchase_Value (tens to hundreds of thousands), EMI (thousands). Without standardization, large-magnitude features (EMI, Purchase_Value) would dominate distance-based models (k-NN, clustering) or slow gradient-based convergence, purely due to scale — not actual importance. Standardizing to mean 0, std 1 ensures every feature contributes fairly.

Q3. Data Transformation & Discretization in Education Analytics (L3, L4, L5)

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good

S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

- Apply **3 bins** for Study_Hours and Test_Score
- Calculate bin width and define intervals

B. Equal-Frequency Binning

- Divide data into **3 bins with equal records**
- Assign values accordingly

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High
- Test_Score → Poor / Average / Excellent

SOLUTION:

A. Equal-Width Binning (3 bins)

Study_Hours (min=2, max=11, range=9, **bin width = $9/3 = 3$**):

Bin	Interval	Students
Low	[2, 5)	S1(2), S2(3), S3(4)
Medium	[5, 8)	S4(5), S5(6), S6(7)
High	[8, 11]	S7(8), S8(9), S9(10), S10(11)

Test_Score (min=40, max=100, range=60, **bin width = $60/3 = 20$**):

Bin	Interval	Students
Poor	[40, 60)	S1(40), S2(50), S3(55)
Average	[60, 80)	S4(65), S5(75)
Excellent	[80, 100]	S6(85), S7(90), S8(95), S9(98), S10(100)

B. Equal-Frequency Binning (3 bins, ~equal record counts: 4-3-3)

Bin	Study_Hours	Test_Score
Bin 1 (S1–S4)	2, 3, 4, 5	40, 50, 55, 65
Bin 2 (S5–S7)	6, 7, 8	75, 85, 90

Bin 3 (S8–S10)	9, 10, 11	95, 98, 100
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Labels assigned the same way: Bin1 = Low/Poor, Bin2 = Medium/Average, Bin3 = High/Excellent.

C. Concept Hierarchy

- **Study_Hours:** Low (2–5 hrs) → Medium (6–8 hrs) → High (9–11 hrs)
- **Test_Score:** Poor (40–59) → Average (60–79) → Excellent (80–100)

This hierarchy converts continuous values into interpretable categories aligned with the "Performance" labels already in the dataset, improving decision-making transparency for non-technical stakeholders.

Q 4. Use the two methods below to normalize the following group of data:

200, 300, 400, 600, 1000

(a) min-max normalization by setting min = 0 and max = 1

(b) z-score normalization

SOLUTION:

Q4. Normalize: 200, 300, 400, 600, 1000

(a) Min-Max Normalization [0, 1]

$X' = (X - X_{min}) / (X_{max} - X_{min})$, $X_{min}=200$, $X_{max}=1000$, range=800

X X'

200 $(200-200)/800 = 0$

300 $(300-200)/800 = 0.125$

400 $(400-200)/800 = 0.25$

600 $(600-200)/800 = 0.5$

1000 $(1000-200)/800 = 1.0$

(b) Z-score Normalization

Mean = $(200+300+400+600+1000)/5 = 500$

Variance (population) = $[(300^2+200^2+100^2+100^2+500^2)] / 5 = 400000/5 = 80000$

Std = $\sqrt{80000} \approx 282.84$

X Z = (X-500)/282.84

200 **-1.061**

300 **-0.707**

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600 **+0.354**

1000 **+1.768**

Q 5. Data Transformation in Online Shopping Behavior Analysis (*L3, L4, L5*)
An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.
To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

Sample Dataset: Visit Duration

Customer_ID	Visit_Duration (seconds)
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

Tasks

- A. Equal-Frequency Binning
- i. Divide the dataset into **3 bins** with approximately equal number of records
 - ii. Assign each value to its respective bin
 - iii. Analyse how effectively this method groups short and long visit durations
- B. Z-Score Normalization
- i. Compute the **mean and standard deviation** of visit duration
 - ii. Apply Z-score normalization:
 - iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**
- C. Min-Max Normalization
- i. Apply Min-Max normalization using range **[0,1]**:
 - ii. Transform all values accordingly
 - iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

SOLUTION:
Data: 15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600
A. Equal-Frequency Binning (3 bins, sizes 4-3-3)

Bin	Values
-----	--------

Bin 1	15, 30, 45, 60
Bin 2	120, 300, 600
Bin 3	900, 1800, 3600

Analysis: This groups quick page-bounces (Bin 1) separately from moderately engaged visits (Bin 2) and highly engaged sessions (Bin 3). It handles the skew reasonably well for classification purposes, but Bin 3 still spans a 4x range (900–3600s), so granularity at the high end is lost — equal-frequency binning prioritizes equal *sample counts* over equal *value ranges*.

B. Z-Score Normalization

Mean = $7470/10 = 747$

Sum of squared deviations = 11,901,060 $\rightarrow$ Variance = 1,190,106 $\rightarrow$ Std ≈ 1090.92

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15	-0.671
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1800	+0.965
3600	+2.616

Handling extreme values: The 3600s visit produces the largest Z-score (+2.62), correctly flagging it as an outlier relative to the distribution, but z-score does **not bound or cap** it — extreme values remain visibly extreme, which can still skew distance-based clustering algorithms even after normalization.

C. Min-Max Normalization [0, 1]

Xmin=15, Xmax=3600, range=3585.

$$X' = (X-15)/3585$$

X	X'
15	0.000
30	0.004
45	0.008
60	0.013

120	0.029
300	0.080
600	0.163
900	0.247
1800	0.498
3600	1.000

Comparison: Min-Max normalization is *more sensitive to the outlier* than Z-score here — because the single extreme value (3600) anchors the entire scale, most other points (even a substantial 600-second visit) get compressed close to 0 (0.163). Z-score instead expresses each value relative to the mean and spread of the *whole distribution*, giving a more

balanced relative picture, though it too is influenced by the outlier's effect on mean/std. For this heavily right-skewed data, a **log-transform prior to normalization** would produce a more balanced result than either method alone.